
YOU RECYCLE
PLASTIC BOTTLES.
BUT WHAT ABOUT THE PLASTIC
YOU'RE WEARING?
PolyCycle SEE
POLYESTER'S HIDDEN IDENTITY
Know Your Fabric. Rethink Your Waste.
100% POLYESTER
Signature (Polyester)
1400
1900
2400
TIER 2 • PROPOSED MULTIMODAL EXPERIMENT
PROJECT CREATOR & DEVELOPER: NICHELLE GRAF
Computer Vision ITAI 1378 • Houston City College • Fall 2026

WE THROW IT AWAY.
BUT IT DOESN'T JUST GO AWAY.

THE PROBLEM: Millions of tons of discarded textiles enter landfills every year, which includes mass amounts of synthetic clothing. These fibers resist biodegradation, just like plastic bottles and packaging. Polyester is a petroleum-derived plastic that can persist in the environment for many years.

WHY IT MATTERS: Synthetic microfibers enter waterways and ecosystems. Textile production and disposal contribute to greenhouse gas emissions, while persistent synthetic fibers and microplastic pollution place additional pressure on ecosystems. Mass disposal consumes valuable land and resources. Limited textile recycling options and complex fiber blends compound the waste problem.

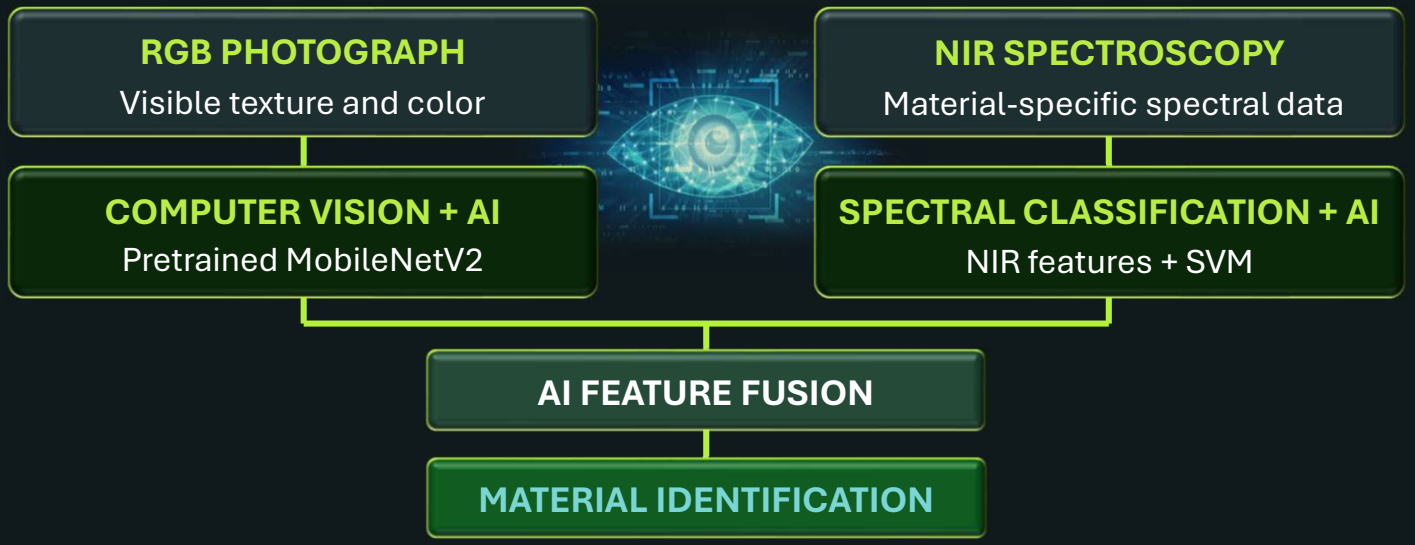
WHO CARES? ALL OF US. Textile waste is a shared global concern. Its consequences extend beyond consumers and the textile industry, affecting communities, human well-being, wildlife, marine life, and the ecosystems on which we all depend.

Sources: EPA, 2018 MSW estimates; GAO-25-107165 (2024).

FROM IDENTIFICATION TO INFORMED ACTION

THE GAP: Consumers may not recognize PET-based fibers or know whether a recycling route exists.

PROPOSED SOLUTION: PolyCycleSEE explores how computer vision, near-infrared (NIR) spectroscopy, and AI can work together to identify textile composition and support informed consumer decisions.



FUTURE CONSUMER EXPERIENCE: IDENTIFY → UNDERSTAND → ACT

**The semester deliverable is a notebook proof of concept—not a sensor-equipped phone app.*

04 / TECHNICAL APPROACH

TWO WAYS OF SEEING.
ONE RESEARCH QUESTION.

RGB IMAGE

Technique:
image classification + visual features

Pretrained MobileNetV2 (CNN)

Frozen feature extraction • TensorFlow/Keras

NIR SPECTROSCOPY

Technique:
spectral classification

Support Vector Machine (SVM)

Train on labeled spectra • scikit-learn

MULTIMODAL FUSION

Technique:
combine RGB + NIR features

Feature-fusion Classifier

Train combined classifier • scikit-learn

WHY THIS APPROACH?

RGB captures visible texture and color; NIR provides complementary material-specific evidence. Together, they may improve polyester/blend identification.

71 FABRICS. TWO TYPES OF EVIDENCE.

71 TEXTILE SAMPLES

RGB photographs + NIR spectral measurements



8 LABELED 100% POLYESTER → 62 WITH RECORDED POLYESTER CONTENT

CAUTIONS: 71 independent samples | class imbalance | one filename mismatch | labels not laboratory-verified.

SOURCES: OpenTextile-NIR (2026) · Zenodo record 18269172
LABELS: Supplier-reported fiber percentages (including blends) **Illustrative textile textures.*

DON'T JUST CLASSIFY.
TEST WHAT ADDS VALUE.

TARGET	F1 ≥ 0.70	Preliminary goal; not a claimed result
COMPARE	RGB vs NIR vs FUSION	Use consistent sample-level splits
REPORT	F1 • precision • recall	Confusion matrices + inference ≤ 2 sec/sample

SUCCESS = working, reproducible computer vision demonstration + candid evaluation of limitations.

BUILD FIRST. IMPROVE WITH EVIDENCE.

W10	BLUEPRINT	Scope + GitHub + proposal
W11	FIRST WORkING DEMO	Pretrained model runs on sample images
W12–13	MAKE IT YOURS	Prepare RGB/NIR data • Train models • Test fusion
W14	IMPROVE & MEASURE	Evaluate, compare, document limitations
W15	PACKAGE & PRESENT	Demo video, final slides, README/report

Proof of concept precedes major training or full hyperspectral downloads.

AMBITIOUS VISION.
PRACTICAL PLAN B.

LIMITED LABELED SAMPLES	Transfer learning; report uncertainty; split by physical sample
RGB MAY MISS FIBER IDENTITY	Demonstrate image feature pipeline; use NIR as complementary modality
MULTIMODAL INTEGRATION DELAYS	Deliver working RGB notebook; document NIR integration as extension

Google Colab • TensorFlow/Keras • scikit-learn • OpenTextile-NIR • GitHub

FREE / OPEN SOURCE